

Week 6 Project Summary

HealthConnect Project – Data Analytics Track

Intern: Hudu Yusuf Ibrahim

Pod: HC Pod 8

Week: 6

1. Planned Objectives

- Validate the strongest Week 5 findings using statistical methods.
- Measure association strength using Chi-square tests and Cramér's V.
- Refine KPIs and improve the Power BI dashboard.
- Complete the required cross-track collaboration with the Data Science track.

2. Work Completed

- Conducted statistical validation using Chi-square tests.
- Calculated Cramér's V effect sizes.
- Validated findings across appointment type, reminder status, and distance bands.
- Promoted the combined Lead Time $\times$ Previous No-Show segment into a formal KPI.
- Refined the Power BI dashboard.
- Completed cross-track collaboration with Data Science interns.

3. Key Findings

- Booking lead time remained the strongest individual factor.
- Previous no-show history was also statistically significant.
- The combined segment (Lead Time $\times$ Previous No-Show) produced the strongest signal.
- Data Science modelling confirmed the importance of booking lead days and previous no-shows.

4. Cross-Track Collaboration

Cross-track collaboration was completed with members of the Data Science track, including Donald Nwachukwu and Fatimah Odumuyiwa.

Key contributions received:

- Logistic Regression achieved the best model performance across the reported modelling evaluations, with ROC-AUC values of approximately 0.67–0.68.

- 'booking_lead_days' and 'previous_no_shows' were confirmed as important features in the predictive modelling analysis.
- The combined Lead Time $\times$ Previous No-Show feature provided additional insight into high-risk appointment segments.
- Distance showed descriptive importance in the Data Analytics analysis but provided limited additional predictive value in the reported model setup after other variables were considered.
- An unexpected finding from Data Science's error analysis was that appointments the model missed (false negatives) were more likely to have received a reminder than the No-Show cases it correctly identified. This suggests that reminder status alone may not be sufficient to identify all No-Show cases and should be investigated further in future modelling.

A small methodological difference was also identified:

- Data Analytics highest-risk segment: 64.1%
- Data Science replication: 70.5%

The difference was attributed to a methodological distinction in the denominator: the Data Analytics calculation included all appointment outcomes, including Cancelled appointments, while the Data Science calculation excluded Cancelled appointments.

This difference highlighted the importance of clearly defining the denominator when calculating and comparing No-Show rates. Despite the difference in percentages, both analyses identified the same overall pattern: No-Show rates were highest among appointments with long booking lead times and previous No-Show history.

5. Challenges Encountered

- Initial difficulty obtaining responses from the Data Science track within HC Pod 8.
- Cross-track collaboration was completed later in the week after additional outreach.
- Care was required to interpret statistical significance separately from practical effect size.

6. Important Decisions

- Promoted the combined Lead Time $\times$ Previous No-Show segment to KPI status.
- Retained appointment type and gender as dashboard filters rather than core KPIs.
- Focused recommendations on validated findings rather than descriptive observations alone.

7. Changes From Week 5

- Added statistical validation.
- Added effect size measurements.
- Refined KPIs.
- Improved dashboard explanations.
- Incorporated Data Science feedback.

8. Remaining Work

- Further collaboration with Data Science on predictive modelling.
- Possible dashboard enhancements.
- Integration of predictive outputs into business recommendations.

9. Proposed Focus for Week 7

- Support predictive model interpretation.
- Integrate analytics findings with Data Science outputs.
- Improve decision-support recommendations for appointment management.